Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

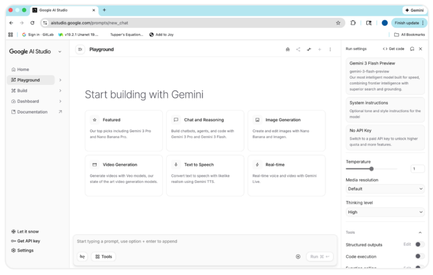
Next item

Activity: Feature discovery

Overview

In this activity, you will explore the powerful features of Google AI Studio by navigating the interface and getting a start to using the platform. You will create a project and an API Key to use in later activities. You get some experience with prompting and will see some of the initial features like text generation and image generation.

AI Studio home page



Scenario

Imagine you are a new developer on a team that wants to use AI to build innovative applications. Your manager has asked you to explore Google AI Studio and report back on its most useful features for the team's upcoming projects. To do this, you will complete five short, hands-on tasks. You will record your findings in an Exploration Log. This log will help your manager and team decide which AI features to focus on for future development.

Before you begin: Materials

What you'll need:

- A Google account to access AI Studio.
- Your Exploration log template. This is a document where you'll record your findings for each mission.

Exploration Log Template
DOCX File

Instructions: You will record your findings and answer the final reflection questions in this document.

What to expect

- **Part 1: Setup and orientation:** Get your API key and get familiar with the AI Studio interface.
- **Part 2: Your five missions:** Complete five hands-on experiments.
- **Part 3: Reflection and submission:** Review your work and answer reflection questions.

Steps

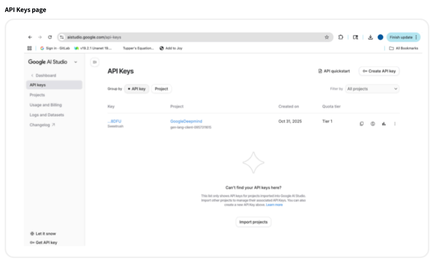
Part 1: Setup and orientation

First, you will get your API key and explore the AI Studio workspace.

Get your API key

1. Navigate to **Google AI Studio** and sign in with your Google account.
2. In the left-hand menu, click **Get API key**.
3. Follow the instructions to **Create API key in new project**.
4. Copy your new key and save it in a secure location on your computer, like a password manager. You will need this for coding projects later.
   - **Important security note:** Treat your API key like a password. Never share it publicly or save it directly in your code where others might see it.
5. Take Note of the Project name as well.

API Keys page



1. You just created a new API key in Google AI Studio. What is the best way to store this key so it stays secure and can be used later for coding projects? Select the best answer. 1 / 1 point

   - (●) Save it in a secure location like a password manager.
   - ( ) Paste it directly into your source code so it's always available.
   - ( ) Add it to your project README so teammates can find it easily.
   - ( ) Email it to yourself for easy access later.

   Nice work

   Correct! Treat your API key like a password—store it in a secure place such as a password manager or secrets vault.

Part 2: Your tasks

Now, you will complete four tasks. For each one, use AI Studio to create a prompt, test the result, and fill out the matching section in your **Exploration Log**.

Task 1: Create a long poem about time and life

- **Goal:** Create a high-quality long-form poem using a text prompt.
- **Your task:** In AI Studio, create a new **Chat prompt**. Write a prompt that generates a long poem exploring time, life, change, memory, and meaning. Ask for a clear structure (stanzas, recurring motif, vivid imagery).
- **Example prompt:** *Act as a literary poet. Write a long poem (at least 40 lines) about time and life. Use vivid imagery, a reflective tone, and a repeating motif (like a clock, ocean tide, or seasons). Include 5–7 stanzas with a strong ending that feels hopeful but honest.*
- **In your exploration log:** Document the feature you used, the output quality (imagery, coherence, length), a possible application (creative writing practice, speech/opening for a talk), and any limitations (repetition, tone drift, clichés).

Task 2: Create an image of a valley of flowers and trees

- **Goal:** Generate a detailed, visually appealing landscape image from an image prompt.
- **Your task:** In AI Studio, create a new Image prompt. Write a prompt that describes a valley filled with wildflowers and trees, including lighting, season, perspective, and style.
- **Example prompt:** *Generate a high-resolution landscape image of a wide valley covered in colorful wildflowers, with scattered oak and pine trees, golden-hour sunlight, soft mist in the distance, and mountains on the horizon. Cinematic composition, ultra-detailed, natural colors, 16:9 aspect ratio.*
- **In your exploration log:** Document the feature you used, the output quality (detail, realism, composition), a possible application (wallpaper, mood board, concept art), and any limitations (strange textures, inconsistent lighting, artifacts).

Task 3: Create a 3-paragraph story about a brave explorer

- **Goal:** Produce a short narrative with strong structure and character using a text prompt.
- **Your task:** In AI Studio, create a new **Chat prompt**. Write a prompt that generates exactly 3 paragraphs with a beginning, middle, and end, featuring a brave explorer facing a meaningful challenge.
- **Example prompt:** *Write a story in exactly 3 paragraphs about a brave explorer who discovers something unexpected. Keep it vivid and emotional, with a clear beginning, conflict, and resolution. Aim for an adventurous tone and end with a memorable final sentence.*
- **In your exploration log:** Document the feature you used, the output quality (clarity, pacing, emotional impact), a possible application (blog/story prompt, classroom exercise), and any limitations (generic plot, weak ending, inconsistent voice).

Task 4: Create an image of a jungle with many animals

- **Goal:** Generate a complex scene image with multiple subjects while maintaining visual coherence.
- **Your task:** In AI Studio, create a new **Image prompt**. Write a prompt describing a dense jungle environment with many animals, specifying composition, animals present, lighting, and style to reduce randomness.
- **Example prompt:** *Create a detailed jungle scene in daylight with lush green foliage, sun rays filtering through the canopy, and a variety of animals visible: a toucan on a branch, a monkey in the midground, a jaguar partially hidden in tall plants, colorful frogs near a stream, and butterflies in the foreground. High detail, realistic style, balanced composition, 16:9 aspect ratio.*
- **In your exploration log:** Document the feature you used, the output quality (animal clarity, scene consistency, realism), a possible application (children's book illustration, educational poster), and any limitations (animals blending into background, incorrect anatomy, overcrowding).

2. You want AI Studio to generate a detailed landscape image of a valley filled with flowers and trees. Which feature should you use to complete this task? Select the best answer. 1 / 1 point

   - ( ) A Video prompt
   - (●) An Image prompt
   - ( ) A Chat prompt
   - ( ) A Structured prompt

   Nice work

   Correct! An Image prompt is designed for generating images from your scene description.

3. You're generating a jungle image that includes many animals, and you want the result to look coherent and not random. What is the best way to improve the prompt? Select the best answer. 1 / 1 point

   - ( ) Use a Chat prompt instead of an Image prompt.
   - ( ) Describe the jungle environment and avoid listing animals.
   - (●) Specify the animals, their positions (foreground/midground/background), lighting, and style.
   - ( ) Keep the prompt short and vague so the model can be more creative.

   Nice work

   Correct! Adding composition details and constraints helps the model place many subjects more consistently.

4. You are completing Task 1 and want to generate a 40-line poem about time. While the model produces a good poem, you notice the tone becomes repetitive in the middle stanzas. According to your Exploration Log task, how should you document this observation? Select the best answer. 1 / 1 point

   - (●) Record it as a "Limitation encountered".
   - ( ) Ignore it since the poem was at least 40 lines long.
   - ( ) Record it as a "Potential real-world application".
   - ( ) Record it as "Video Generation".

   Nice work

   Correct! Observations like tone drift, clichés, or repetition are considered limitations of the model's output that you should track in your log.

5. Your manager asks which AI Studio feature is best for creating a short, 3-paragraph narrative about a brave explorer (Task 3). Based on your activity instructions, which tool should you recommend? Select the best answer. 1 / 1 point

   - ( ) Video prompt
   - ( ) Image prompt
   - (●) Chat prompt
   - ( ) API Key menu

   Nice work

   Correct! A Chat prompt is the appropriate tool for generating long-form text, poems, and structured stories.

Part 3: Reflect on your discoveries

It is time to prepare your "report" for your manager. Review your completed Exploration Log and answer the following questions in the "Reflection" section of your log document. This helps you think critically about what you have learned.

Write your reflection:

- **Most exciting feature:** Which task was the most interesting or surprising to you? Which one are you most excited to use in a future project, and why?
- **Biggest challenge:** In which task did the model have the most trouble or give the weakest result? What do you think caused this problem?

After you finish your reflection, your Exploration Log is complete.

Congratulations!

You have successfully explored Google AI Studio and created five unique prototypes.

Success checklist

- You have successfully created and securely saved a Gemini API key.
- You have completed all missions and filled out your Exploration Log.
- You have completed the reflection questions based on your discoveries.

Key points

- AI Studio is a tool for quickly testing AI ideas before writing code.
- Different prompt types (Image or Chat) are used for different tasks.